LIFTING MAPS BETWEEN THE TATE RESOLUTION AND KOSZUL TOWERS

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Editorial note. 'The map phi_2 exists and is compatible with the differential and the divided powers by the universal [property] of the divided power polynomial algebra' – the universal-property invocation is in-chapter template (lemma-divided-power-polynomial-algebra); no other omission markers.

Allowed standard prerequisites (editorial).

- Tate resolution construction. lemma-tate-resolution (same chapter, named). Supplies the inductive skeleton this proof reuses ('exactly as in the statement and proof of Lemma lemma-tate-resolution').
- Koszul transition selection. more-algebra-lemma-sequence-Koszul-complexes (cross-chapter, named). Chooses the tower n = n_0 < n_1 < ... < n_r = N with zero maps on positive homology, the spine of the construction.

Source reference. Original definition 09PJ: sources/stacks-dpa/pinned-original-source.tex, source lines 1300–1300. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09PI: sources/stacks-dpa/pinned-original-source.tex, source lines 1241–1241. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09PH: sources/stacks-dpa/pinned-original-source.tex, source lines 1221–1221. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09PN: sources/stacks-dpa/pinned-original-source.tex, source lines 1424–1424. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0921: focused-seeds/geometry/S9-dpa-lift-tate-to-koszul/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In the situation above, if R is Noetherian, then for every n there exists an $N \geq n$ and maps*

$$K_N \to A \to R/(f_1^N, \ldots, f_r^N) \quad \text{and} \quad A \to K_n$$

with the following properties

(1) *(A, d, γ) is as in Definition 09PJ,*
(2) *$A \to R/(f_1^N, \ldots, f_r^N)$ is a quasi-isomorphism,*
(3) *the composition $K_N \to A \to R/(f_1^N, \ldots, f_r^N)$ is the canonical map,*
(4) *the composition $K_N \to A \to K_n$ is the transition map,*
(5) *$A_0 = R \to R/(f_1^N, \ldots, f_r^N)$ is the canonical surjection,*
(6) *A is a graded divided power polynomial algebra over R with finitely many generators in each degree, and*
(7) *$A \to K_n$ is a homomorphism of differential graded R-algebras compatible with divided powers which induces the canonical map $R/(f_1^N, \ldots, f_r^N) \to R/(f_1^n, \ldots, f_r^n)$ on homology in degree 0.*

Condition (4) means that A is constructed out of A_0 by successively adjoining a finite set of variables T in each degree > 0 as in Example 09PH or 09PI.

Proof. Fix n. If $r = 0$, then we can just pick $A = R$. Assume $r > 0$. By More on Algebra, Lemma 0921 (translated into the language of chain complexes) we can choose

$$n_r > n_{r-1} > \ldots > n_1 > n_0 = n$$

such that the transition maps $K_{n_{i+1}} \to K_{n_i}$ on Koszul algebras (see above) induce the zero map on homology in degrees > 0. We will prove the lemma with $N = n_r$.

We will construct A exactly as in the statement and proof of Lemma 09PN. Thus we will have

$$A = \operatorname{colim} A(m), \quad \text{and} \quad A(0) \to A(1) \to A(2) \to \ldots \to R/(f_1^N, \ldots, f_r^N)$$

This will immediately give us properties (1), (2), (5), and (6). To finish the proof we will construct the R-algebra maps $K_N \to A \to K_n$. To do this we will construct

(1) an isomorphism $A(1) \to K_N = K_{n_r}$,
(2) a map $A(2) \to K_{n_{r-1}}$,
(3) $\ldots$
(4) a map $A(r) \to K_{n_1}$,
(5) a map $A(r+1) \to K_{n_0} = K_n$, and
(6) a map $A \to K_n$.

In each of these steps the map constructed will be between differential graded algebras compatibly endowed with divided powers and each of the maps will be compatible with the previous one via the transition maps between the Koszul algebras and each of the maps will induce the obvious canonical map on homology in degree 0.

Recall that $A(0) = R$. For $m = 1$, the proof of Lemma 09PN chooses $A(1) = R\langle T_1, \ldots, T_r \rangle$ with T_i of degree 1 and with $\mathrm{d}(T_i) = f_i^N$. Namely, the f_i^N are

generators of the kernel of $A(0) \to R/(f_1^N, \ldots, f_r^N)$. Thus for $A(1) \to K_N = K_{n_r}$ we use the map

$$\varphi_1 : A(1) \longrightarrow K_{n_r}, \quad T_i \longmapsto \xi_i$$

which is an isomorphism.

For $m = 2$, the construction in the proof of Lemma 09PN chooses generators $e_1, \ldots, e_t \in \mathrm{Ker}(\mathrm{d} : A(1)_1 \to A(1)_0)$. The construction proceeds by taking $A(2) = A(1)\langle T_1, \ldots, T_t\rangle$ as a divided power polynomial algebra with T_i of degree 2 and with $\mathrm{d}(T_i) = e_i$. Since $\varphi_1(e_i)$ is a cocycle in K_{n_r} we see that its image in $K_{n_{r-1}}$ is a coboundary by our choice of n_r and n_{r-1} above. Hence we can construct the following commutative diagram

$$\begin{array}{ccc} A(1) & \xrightarrow[\varphi_1]{} & K_{n_r} \\ \downarrow & & \downarrow \\ A(2) & \xrightarrow{\varphi_2} & K_{n_{r-1}} \end{array}$$

by sending T_i to an element in degree 2 whose boundary is the image of $\varphi_1(e_i)$. The map φ_2 exists and is compatible with the differential and the divided powers by the universal of the divided power polynomial algebra.

The algebra $A(m)$ and the map $\varphi_m : A(m) \to K_{n_{r+1-m}}$ are constructed in exactly the same manner for $m = 2, \ldots, r$.

Given the map $A(r) \to K_{n_1}$ we see that the composition $H_r(A(r)) \to H_r(K_{n_1}) \to H_r(K_{n_0}) \subset (K_{n_0})_r$ is zero, hence we can extend this to $A(r+1) \to K_{n_0} = K_n$ by sending the new polynomial generators of $A(r+1)$ to zero.

Having constructed $A(r+1) \to K_{n_0} = K_n$ we can simply extend to $A(r+2), A(r+3), \ldots$ in the only possible way by sending the new polynomial generators to zero. This finishes the proof. □